

Harsh Shukla

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EDUCATION

McMaster University

Bachelor of Engineering in Software Engineering (Co-op), GPA: 3.9/4.0

Hamilton, ON

Sep 2023 – Apr 2028 (Expected)

EXPERIENCE

Software Developer

Intact Financial Corporation

Sept 2025 – Present

Toronto, Canada

- Engineered core components of the Enterprise Content Management (ECM) service using Java Spring Boot, optimizing document retrieval speeds by 25% for internal stakeholders.
- Architected a cloud-native Claims Management System using AWS Lambda, API Gateway, and DynamoDB, migrating legacy data to reduce query latency by 45% and operational costs by 30%.
- Resolved 15+ critical performance bottlenecks by implementing MongoDB index optimizations and rigorous JUnit testing, increasing system scalability by 30%.

Full Stack Software Engineering Intern

Laurette Software

Apr 2025 – Aug 2025

Remote

- Deployed a scalable school directory on AWS and Vercel, serving 100+ active users by architecting a normalized PostgreSQL schema for complex relationship mapping.
- Secured user data by implementing Auth0 authentication with JWT strategy and role-based access control, maintaining 99.9% system availability.

Software Developer, Exoskeleton Team

McMaster University

Dec 2024 – Aug 2025

Hamilton, ON

- Developed a high-frequency data pipeline for 10+ inertial sensors using C++ and ROS, slashing latency by 60% to enable real-time gait control.
- Implemented an Extended Kalman Filter for multi-sensor fusion, improving positional tracking accuracy by 35% during dynamic movement.
- Deployed AI-driven gait prediction models in Python, achieving 92% accuracy in stride phase detection for adaptive robotic assistance.

PROJECTS

Badge: AI-Powered Event Networking Platform (UofTHacks 13 Winner)

Jan 2026

- Implemented an AI-driven identity extraction and matching system that parses resumes using **Gemini LLMs**, generates **768-dimensional vector embeddings** with **Snowflake Cortex**, and computes compatible connections.
- Engineered a pipeline using **ElevenLabs** and the **Gemini API** to analyze live conversations, extract sentiment and action items, to power a personalized networking coach with persistent memory via **Backboard.io**.
- Architected a scalable, real-time platform using **Convex** serverless functions and database sync, **Clerk** authentication for secure identity management, and a modular design to support live interactions and AI workflows.

Astra: Engineering Intelligence & RAG System

Aug 2025

- Architected a **Retrieval-Augmented Generation (RAG)** pipeline using **Pinecone** and the **OpenAI API** to synthesize complex engineering documentation, achieving **98%** factual accuracy.
- Engineered a multi-agent orchestration layer via **LangGraph** that autonomously identifies knowledge gaps, adapts response strategies, and summarizes technical information for end-users.
- Designed an **observability-focused frontend and analytics layer** with role-based views, custom metrics, and interaction telemetry to monitor agent performance, user comprehension trends, and system reliability at scale.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frameworks and Tools: React, Node.js, Docker, Kubernetes, Firebase, Azure, ROS, Git, GitHub Actions, Jenkins, Postman, UML, Auth0, JWT

Concepts: Data Structures and Algorithms, Object-Oriented Programming, RESTful APIs, Microservices, Agile and Scrum, CI/CD